

6 Further sequences and series	6.1	First and second order recurrence relations.	Equations of the form $u_{n+1} + f(n)u_n = g(n)$ and $u_{n+2} + f(n)u_{n+1} + g(n)u_n = h(n)$
	6.2	The solution of recurrence relations to obtain closed forms.	Students should be able to solve relations such as: $u_{n+1} - 5u_n = 8, u_1 = 1,$ $2u_{n+2} + 7u_{n+1} - 15u_n = 6, u_1 = 10, u_2 = -17$ The terms, particular solution, complementary function and auxiliary equation should be known. Use of recurrence relations to model applications, e.g. population growth.
	6.3	Proof by induction of closed forms.	For example if $u_{n+1} - 3u_n = 4$ with $u_1 = 1$ prove by mathematical induction that $u_n = 3^n - 2$